IAT New Test 13

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

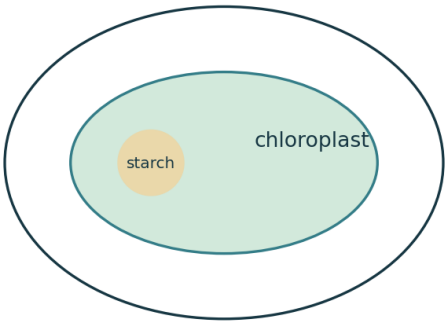
Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 A microscopic specimen in the figure has chlorophyll a and b and stores starch within a chloroplast. Which algal group is most consistent with these linked traits?

Algal cell pigment and reserve

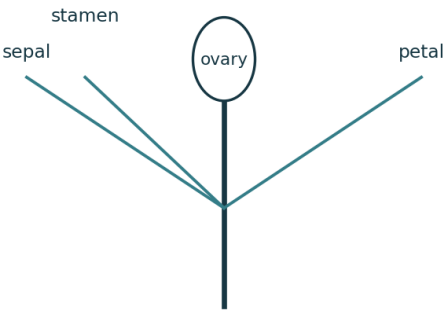


Pigments extracted: chlorophyll a and b

A) Chlorophyceae
B) Phaeophyceae
C) Rhodophyceae
D) Cyanobacteria

Q2 The flower diagram shows a superior ovary with sepals, petals and stamens attached below it. Which position is represented?

Longitudinal flower section



Whorl insertions below ovary

A) Epigynous
B) Hypogynous
C) Perigynous
D) Inferior ovary with epigynous insertion

Q3 A eukaryotic messenger RNA leaves the nucleus and is translated into a secreted protein. Where does translation begin and where does elongation proceed after the signal peptide is recognised?
A) Inside nucleus, then lysosome
B) Mitochondrion, then cytosol
C) Cytosolic ribosome, then ribosome associated with rough ER
D) Golgi lumen, then plasma membrane

Q4 In the Calvin cycle, three CO_2 molecules are fixed. How many molecules of the three-carbon triose phosphate leave the cycle as the net carbohydrate product while the rest regenerate RuBP?
A) Three
B) Two
C) Six
D) One

Q5 A contracting sarcomere shortens while thin and thick filaments keep their own lengths. Which pair of band changes is expected?
A) I band narrows, A band remains approximately constant
B) A band narrows, I band remains constant
C) Both bands narrow in proportion
D) Both bands remain constant

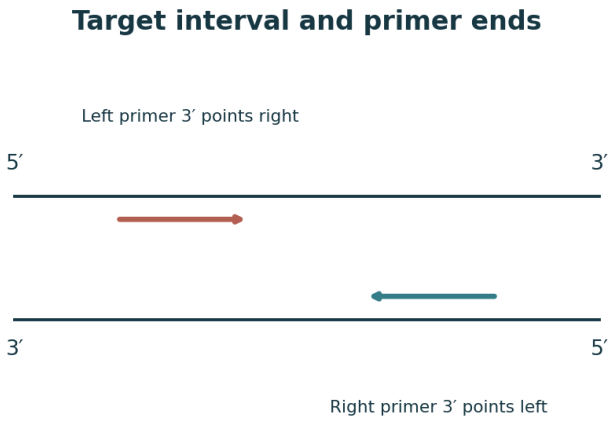
Q6 In an angiosperm cross, the maternal sporophyte is 2n=24 and paternal pollen-producing plant is also 2n=24. Assuming normal haploid gametes, what are embryo and endosperm chromosome numbers after double fertilisation?
A) 24,24
B) 24,36
C) 36,24
D) 12,36

Q7 In a replication fork moving right, the upper parental template is read 3′→5′ toward the fork. Which statement describes new synthesis on that template?
A) The strand is made 3′→5′ continuously
B) The strand is made 5′→3′ only as short fragments away from fork
C) The leading strand grows 5′→3′ toward the fork
D) No DNA can be made on this template

Q8 Two purified immune-cell populations are tested with the same antigen and all other culture conditions held fixed. B cells alone make little specific antibody; helper T cells alone make none; the combined culture makes much more. Preventing helper-cell signalling in the combined culture reduces antibody production, while supplying ready-made antibody still neutralises the antigen. Which inference is best supported?
A) Helper T cells themselves are the only antibody-secreting cells
B) The neutralising antibody can be produced without any B-cell lineage
C) The blocking treatment necessarily destroys antigen
D) Helper T cells promote the B-cell antibody response, while ready-made antibody acts downstream of that cooperation

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Q9 A linear double-stranded DNA target is shown with top strand 5′ at left and 3′ at right. Which primer orientation permits amplification of the interval between the two inward-facing sites?



A) Left forward primer extends right; right reverse primer extends left
B) Both primers extend right
C) Both primers extend left
D) Both primers anneal with 3′ ends pointing outward

Q10 A grassland fixes 1400 kJ m^{-2} day^{-1} as gross primary productivity and autotrophs respire 500 kJ m^{-2} day^{-1}. Herbivores assimilate 120 kJ m^{-2} day^{-1}. What is plant net primary productivity?
A) 120
B) 900
C) 1020
D) 1900

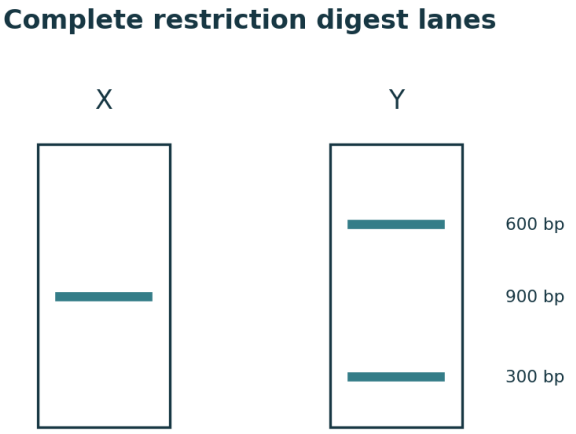
Q11 In an inducible bacterial operon, a functional repressor blocks transcription without inducer. The operator DNA is mutated so that the repressor can no longer bind, while coding genes remain intact. What happens in absence of inducer?
A) Transcription necessarily stops permanently
B) Only translation, not transcription, changes
C) Constitutive transcription becomes possible
D) The repressor protein itself is deleted

Q12 An mRNA reading frame begins AUG–GAA–UAA. If translation begins at the first AUG, how many amino acids are incorporated before termination?
A) One
B) Three
C) Four
D) Two

Q13 Two patients have similarly low circulating thyroid hormone. Patient P has high TSH; patient Q has low TSH. Which distinction is most consistent with intact negative feedback in P and impaired pituitary drive in Q?
A) Neither can have thyroid disease
B) P has pituitary failure, Q has primary thyroid failure
C) Both necessarily have iodine excess
D) P has primary thyroid failure, Q may have secondary pituitary failure

Q14 A diploid cell has 2n=8 and 2C DNA in G1. At metaphase after replication but before sister chromatids separate, how many chromosomes and how much DNA does it contain?
A) 16 chromosomes, 4C
B) 8 chromosomes, 4C
C) 8 chromosomes, 2C
D) 16 chromosomes, 2C

Q15 A DNA digest gel has a 900 bp band in lane X and 600 bp plus 300 bp bands in lane Y; full sample DNA lengths are 900 bp. Which statement follows from band positions alone?



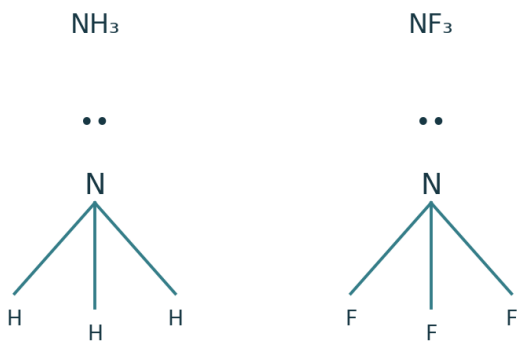
A) Lane Y contains twice as much DNA as lane X
B) Lane X is necessarily heterozygous
C) Lane Y DNA was cut at one internal position, while X was not cut at that site
D) The 600 bp fragment must fluoresce twice as brightly as the 300 bp band

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Chemistry

Q16 The structures show NH_3 and NF_3 with three bonds and one lone pair on central nitrogen. Which has the larger H–N–H or F–N–F bond angle, respectively?

Nitrogen bond-pair directions



A) NH_3, because N–H bond pairs stay nearer N than N–F bond pairs
B) NF_3, because F is larger than H
C) Equal, because both have four domains
D) NF_3, because lone pairs vanish in polar bonds

Q17 In an octahedral d^6 complex, a strong-field ligand gives a low-spin state. How many unpaired electrons are expected?
A) Four
B) Zero
C) Two
D) Six

Q18 An ion's measured spin-only magnetic moment is $\sqrt{24}$ Bohr magnetons. How many unpaired electrons does the spin-only model assign, and what is the ratio of that number to the count for a $\sqrt{3}$-BM ion?
A) 2 and 2
B) 3 and 3
C) 4 and 4
D) 6 and 6

Q19 Two electrons are proposed in the same atom with identical (n,l,m_l,m_s) quantum numbers. Which rule rules this out even if the orbital energies are equal?
A) Hund's rule only
B) Aufbau principle only
C) Heisenberg uncertainty only
D) Pauli exclusion principle

Q20 Among Be, B, N and O in the second period, which paired first-ionisation-energy comparisons are both correct, and what orbital effects explain them?
A) Be>B due to starting a 2p electron in B; N>O because paired 2p electrons in O repel
B) B>Be due to greater nuclear charge; O>N due to greater nuclear charge
C) Be>B because Be has a filled 2p shell; N>O because N has no p electrons
D) B>Be because B is more metallic; N>O because O has a half-filled p shell

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Q21 A hydrogen–oxygen gas mixture has volume 30.0 mL. After complete reaction and cooling until all product water condenses, 5.0 mL oxygen remains. All gas volumes are compared at the same temperature and pressure; neither initial gas contains water. What was the initial hydrogen volume?
A) 25.0 mL
B) $50/3$ mL
C) 20.0 mL
D) 10.0 mL

Q22 Which acid is stronger in water, CH_3COOH or $ClCH_2COOH$, and why?
A) CH_3COOH because methyl withdraws electrons
B) Equal because both are carboxylic acids
C) $ClCH_2COOH$ because chlorine withdraws electron density inductively and stabilises its conjugate base
D) $ClCH_2COOH$ because it has more resonance structures

Q23 In aqueous nucleophilic addition of HCN to acetone under suitable conditions, which atom of CN^- bonds to the carbonyl carbon in the usual cyanohydrin product?
A) Hydrogen
B) Oxygen
C) Nitrogen
D) Carbon

Q24 A liquid gives H_2 with sodium metal but no CO_2 with aqueous $NaHCO_3$. It is soluble in water and has no carbonyl peak. Which class is best supported?
A) Alcohol
B) Carboxylic acid
C) Aldehyde
D) Hydrocarbon

Q25 Starting with benzene, which order of operations gives predominantly 1-bromo-3-chlorobenzene? Use ordinary electrophilic directing effects, selective nitro-group reduction, and Sandmeyer replacement of an aryl diazonium group by Cl.
A) Brominate; nitrate; reduce NO_2; diazotise then CuCl
B) Nitrate; brominate; reduce NO_2; diazotise at 0–5°C then CuCl
C) Nitrate; reduce NO_2; brominate directly with bromine water; diazotise then CuCl
D) Chlorinate; brominate; nitrate; reduce NO_2

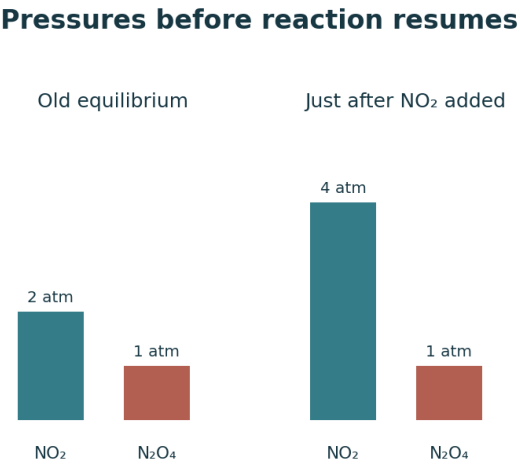
Q26 An amino acid has one amino and one carboxyl group, with no ionisable side chain. At pH near its isoelectric point, which species predominates?
A) $NH_2–CHR–COOH$
B) $NH_3^+–CHR–COOH$
C) $NH_3^+–CHR–COO^-$
D) $NH_2–CHR–COO^{2-}$

Q27 A binary liquid mixture has vapour pressure above its Raoult-law ideal value at intermediate composition. Which pair of mixing tendencies is most consistent with the usual positive-deviation model?
A) Stronger unlike attractions; heat released
B) Stronger unlike attractions; heat absorbed
C) Weaker unlike attractions; heat released
D) Weaker unlike attractions; heat absorbed

Q28 For a strong electrolyte at fixed temperature, dilution decreases specific conductivity κ but increases molar conductivity Λm toward a limiting value. Which statement explains both?
A) Ion concentration per volume falls while conductivity per mole can rise as interactions weaken
B) Ion concentration per volume increases under dilution
C) Each ion loses its charge under dilution
D) Λm must equal κ numerically at every concentration

Q29 A first-order reaction has half-life 20 min. After 60 min, what fraction of initial reactant remains?
A) 1/4
B) 1/32
C) 1/16
D) 1/8

Q30 For $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, the diagram gives an equilibrium mixture with pNO2=2 atm and pN2O4=1 atm, then NO_2 pressure is instantaneously increased to 4 atm at constant T before reaction. Relative to the old equilibrium, which way does reaction initially proceed?



A) Right because NO_2 was added
B) No shift because T is unchanged
C) Left because Qp rises from 4 to 16
D) Right because the product pressure is squared

Mathematics

Q31 A hyperbola has equation $x^2/9 - y^2/16 = 1$. What is the acute angle θ that either asymptote makes with the positive x-axis, through $\tan\theta$?
A) 4/3
B) 3/4
C) 4/5
D) 5/4

Q32 The line $3x + 4y = 12$ meets the axes at A and B. A circle has AB as a diameter. What is the tangent to this circle at the origin?
A) $3x + 4y = 0$
B) $4x + 3y = 0$
C) $4x - 3y = 0$
D) $x + y = 0$

Q33 Let $f(x) = ax^2$ for $x \le 1$ and $f(x) = bx + c$ for $x > 1$. If f is differentiable at $x = 1$ and $f(2) = 5$, what is c?
A) -1
B) 1
C) $-5/3$
D) 5

Q34 Evaluate $\int_0^1 xe^x\,dx$.
A) $e - 1$
B) e
C) 1
D) 0

Q35 Let $f(x) = \sqrt{(4 - x^2)}$ and $g(x) = 1/(x - 1)$. What is the domain of $g \circ f$?
A) $[-2, 2]$
B) $(-2, 2)$
C) $[-2, 2]$ excluding ± 1
D) $[-2, 2]$ excluding $\pm\sqrt{3}$

Q36 A 3×3 determinant D has rows r_1, r_2, r_3. After replacing r_2 by $r_2 + 3r_1$ and interchanging r_1 with r_3, the result is -10. What is D?
A) -30
B) -10
C) 10
D) 30

Q37 In a hypothetical classification experiment, 10% of objects are type X. A test returns positive with probability 0.8 for X and 0.1 for non-X. Two repetitions are independent conditional on the object's type. An object tests positive first and negative second. What is the probability it is X?
A) 8/17
B) 16/97
C) 8/9
D) 1/10

Q38 In $(x+2)^6$, what is the ratio of the coefficient of x^3 to the coefficient of x^4?
A) 2
B) 4/3
C) 3
D) 8/3

Q39 A geometric progression has first term 1 and common ratio 2. Its first n terms sum to 1023. An arithmetic progression has the same n, the same first term and the same last term as those GP terms. By how much does the AP sum exceed the GP sum?
A) 1023
B) 2565
C) 1542
D) 2046

Q40 Let f(x)=sin(kx)/x for nonzero x and f(0)=2. For which real k is f continuous at x=0?
A) −2
B) 0
C) 1
D) 2

Q41 Find the foot of perpendicular from P=(3,4) to the line 3x+4y−10=0.
A) (0,5/2)
B) (6/5,8/5)
C) (9/5,12/5)
D) (3,1/4)

Q42 A path from (0,0) to (4,4) uses only right and up steps. How many paths pass through exactly one of A=(1,2) and B=(3,2)?
A) 60
B) 42
C) 51
D) 9

Q43 A polynomial with real coefficients has 2+i as a root. Which statement must be true?
A) 2−i is also a root
B) −2+i is also a root
C) Its degree is exactly two
D) Its constant term is positive

Q44 For x>0, let $y' = 1 + y/x$ and y(1)=0. What is the area between the solution curve and the line $y = x - 1$ over $1 \le x \le e$?
A) $(4e - e^2 - 1)/4$
B) $(e^2 - 1)/4$
C) $(e^2 + 1)/4$
D) $e - 1$

Q45 If sin θ+cos θ=1/2 for real θ, what is sin 2θ?
A) −3/4
B) −1/2
C) 1/4
D) 3/4

Physics

Q46 A displacement obeys $x=At^2+B\sin(\omega t)$, where x is length and t is time. What are dimensions of A and B, respectively?
A) $L\,T^{-1}$ and $L\,T^{-2}$
B) $L\,T^{-2}$ and L
C) L and T^{-2}
D) $L\,T^{-2}$ and L T

Q47 Two blocks of masses m and 2m are connected by a light string over a frictionless pulley. What is acceleration magnitude when released?
A) g/2
B) g/4
C) g/3
D) 2g/3

Q48 Two wires of equal material and length, with cross-sectional areas A and 2A, are connected in series and pulled by force F. What is extension of the narrower wire divided by extension of the wider one?
A) 1/2
B) 1
C) 4
D) 2

Q49 A metal rod fits exactly between two rigid supports at 20°C. Temperature rises by ΔT while supports stay fixed and the rod remains elastic. What is the magnitude of compressive stress, if its Young modulus is Y and expansion coefficient α?
A) YαΔT
B) YΔT/α
C) αΔT/Y
D) Zero because length is fixed

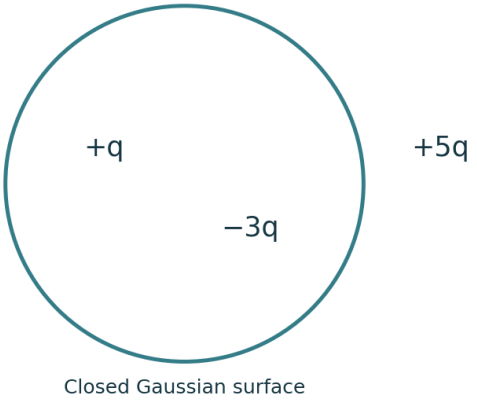
Q50 A diatomic ideal gas has rotations active but vibrations frozen in the listed temperature range. If its measured molar CV is near 5R/2, what is expected molar CP?
A) 3R/2
B) 7R/2
C) 5R/2
D) 9R/2

Q51 At a point on a circular path with constant speed, velocity points east. Which direction does acceleration point if the centre is north of that point?
A) East
B) West
C) North
D) Zero

Q52 A wave on a string reaches a boundary with a much denser string, while the incident pulse is upward. What is the direction of the reflected pulse close to a fixed-end limit?
A) Upward with no phase reversal
B) No reflection
C) Always zero displacement everywhere
D) Downward due to approximate π phase reversal

Q53 A closed spherical Gaussian surface encloses +q and −3q; a +5q charge sits just outside it. What is the total electric flux through the sphere?

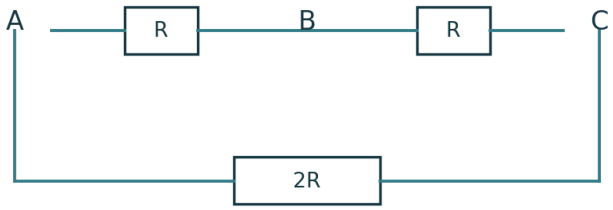
Enclosed versus external charges



A) $-2q/\varepsilon_0$
B) $+3q/\varepsilon_0$
C) $+5q/\varepsilon_0$
D) $-3q/\varepsilon_0$

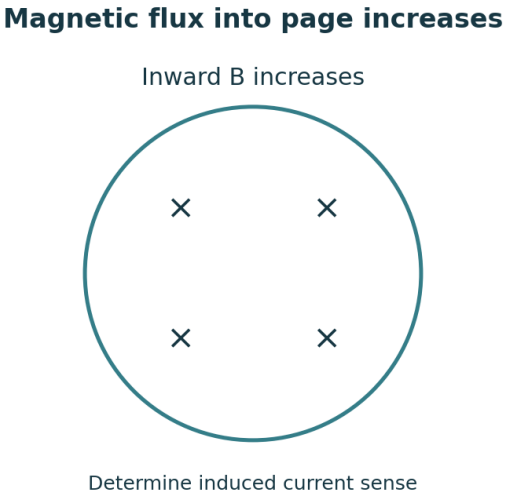
Q54 A network has R between A and B, another R between B and C, and a direct resistor 2R from A to C. Ideal wires make B a distinct node. What is equivalent resistance between A and C?

Two paths between terminals A and C



A) R
B) R/2
C) 2R
D) 4R

Q55 A circular loop is viewed face-on. A magnetic field into the page increases in magnitude. Which direction is induced conventional current?



A) Clockwise, reinforcing inward flux
B) Counterclockwise, producing outward field
C) Clockwise, producing outward field
D) No induced current because field direction stays fixed

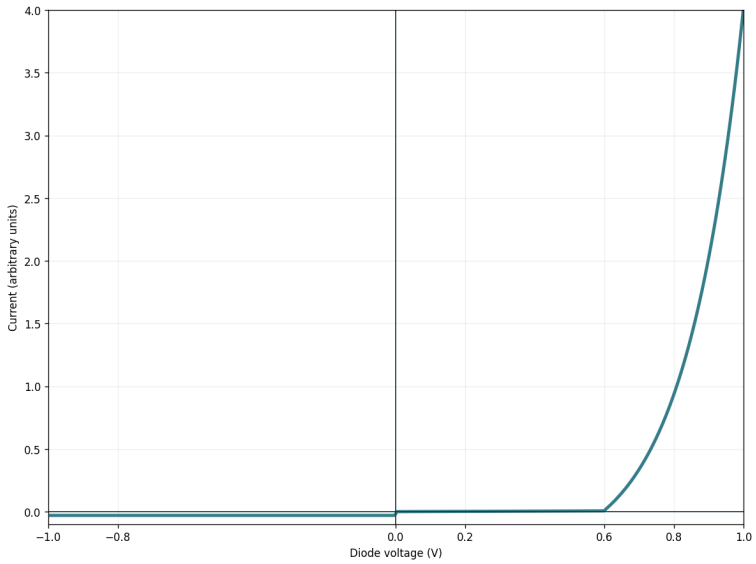
Q56 In a series AC circuit with R=6 Ω, XL=10 Ω and XC=2 Ω at the drive frequency, what is impedance magnitude?
A) 8 Ω
B) 14 Ω
C) 10 Ω
D) 18 Ω

Q57 A thin converging lens of focal length 20 cm forms a real image of an object 30 cm in front of it. A plane mirror is placed exactly at the image plane, perpendicular to the axis. Where do returned rays form the final image after passing back through the lens?
A) At the lens
B) 30 cm on the object side
C) 20 cm on the object side
D) 60 cm on the mirror side

Q58 Two coherent beams interfere with $I_{\text{max}}/I_{\text{min}} = 9$. The weaker beam's intensity is then reduced to one quarter of its original value, with coherence maintained. What is the new visibility $(I_{\text{max}} - I_{\text{min}})/(I_{\text{max}} + I_{\text{min}})$?
A) 4/5
B) 1/2
C) 1/4
D) 8/17

Q59 An electron and a proton each start from rest and are accelerated through the same potential difference V. What is the ratio of their nonrelativistic de Broglie wavelengths λe/λp?
A) √(mp/me)
B) √(me/mp)
C) mp/me
D) 1

Q60 The diode I–V plot has nearly zero forward current up to about 0.6 V, then a sharp rise; reverse current remains small until a large negative voltage. Which operating point best corresponds to ordinary forward conduction?



A) −0.2 V, large current
B) +0.2 V, large current
C) −0.8 V, large negative current
D) +0.8 V, large positive current

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / A algal pigment and reserve product

Answer: (A) Green algae.

Why: Green algae commonly possess chlorophyll a and b and store starch associated with chloroplasts. Pigment plus reserve product is stronger evidence than colour alone because lighting or age can alter appearance.

Why not the others: (B) brown algae have fucoxanthin and laminarin; (C) red algae use phycobilins and floridean starch; (D) are prokaryotic and lack chloroplasts.

Key point: Combine pigment and reserve traits.

Q2 / B flower whorl position

Answer: (B) Hypogynous.

Why: The other whorls arise below a superior ovary, the defining spatial arrangement of a hypogynous flower. A cup-like insertion around, but not above, a free ovary would instead support perigyny.

Why not the others: (A) epigynous insertion places whorls above an inferior ovary; (C) perigynous insertion places them around a cup; (D) explicitly names an inferior ovary contrary to the diagram.

Key point: Locate the insertion relative to ovary base.

Q3 / C ribosome translation context

Answer: (C) Cytosol to rough ER.

Why: Protein synthesis starts on cytosolic ribosomes; a signal sequence directs the translating ribosome to the ER membrane, allowing the nascent secretory chain to enter the ER. The RNA is not translated within the nucleus.

Why not the others: (A) the nucleus processes RNA but does not host the described translation; (B) mitochondria do not initiate this nuclear-encoded secretory route; (D) Golgi modifies cargo after ER synthesis rather than translating mRNA.

Key point: Signal recognition redirects an ongoing translation.

Q4 / D Calvin-cycle carbon tracing

Answer: (D) One net triose.

Given: Three CO_2 add three carbons to the cycle.

Formula/Principle: Six triose phosphates form transiently; five regenerate three RuBP molecules.

Step 1: Six minus five gives one net three-carbon triose.

Why: The six immediate reduction products are not six exported products: most carbon must recycle into the acceptor to keep fixation going. The three fixed carbon atoms leave as one net three-carbon product, while the other five triose units recycle their carbon skeletons to regenerate the acceptor.

Why not the others: (A) counts RuBP molecules; (B) assumes a glucose product from three CO_2; (C) counts all transient trioses.

Key point: Distinguish gross intermediate formation from net export.

Q5 / A skeletal muscle contraction

Answer: (A) I band narrows; A band persists.

Why: Sliding filament overlap increases without shortening the myosin thick filament, whose length sets the A band. The thin-only region at each sarcomere end shrinks, reducing I-band width.

Why not the others: (B) reverses bands; (C) assumes filament shortening; (D) ignores changing overlap.

Key point: Band widths reflect filament lengths and overlap differently.

Q6 / B double-fertilisation ploidy

Answer: (B) 24 and 36.

Given: Egg, each polar nucleus and each sperm have n=12.

Formula/Principle: Embryo=egg+sperm; primary endosperm nucleus=two polar nuclei+second sperm.

Step 1: Embryo 12+12=24; endosperm 12+12+12=36.

Why: The two sperm participate in different fusion events, and central-cell maternal contribution includes two haploid nuclei. Embryo and endosperm therefore have different ploidy even though both fertilisation events use sperm from the same pollen grain.

Why not the others: (A) omits a polar nucleus; (C) swaps tissues; (D) omits sperm contribution to embryo.

Key point: Track each nucleus in each fusion.

Q7 / C molecular replication direction

Answer: (C) Continuous leading synthesis.

Why: Polymerase reads a template 3′→5′ and extends the daughter strand 5′→3′. When the template orientation points into fork movement, extension follows the fork and can be continuous.

Why not the others: (A) reverses polymerase chemistry; (B) describes the lagging configuration; (D) ignores complementary directionality.

Key point: Align template reading direction with fork travel.

Q8 / D Helper-B-cell cooperation with a downstream bypass

Answer: (D) Helper-cell cooperation supports antibody production; supplied antibody bypasses that production step.

Why: Comparing the two isolated populations with their combination shows a cooperative effect rather than antibody secretion by T cells alone. Selectively blocking helper signalling reduces the combined response, supporting its causal contribution under the controlled conditions. Neutralisation by supplied antibody shows that the final antibody-effector step can still function when upstream cell cooperation is bypassed.

Why not the others: (A) contradicts the helper-only culture and B-cell lineage of antibody-secreting plasma cells; (B) is not established by adding already-made antibody; (C) is contradicted by continued neutralisation of the antigen after antibody is supplied.

Key point: Use isolated populations, a targeted block and a downstream bypass to locate an immune function.

Q9 / A PCR primer orientation

Answer: (A) Primers must face each other.

Why: DNA polymerase extends each primer from its 3′ end in the 5′→3′ direction. For exponential amplification of the bounded interval, extension from the left site proceeds right and from the right site proceeds left on opposing templates.

Why not the others: (B) sends both extensions in one direction, so one flank is not copied inward; (C) reverses that common direction but still leaves a flank; (D) points both free 3′ ends away from the interval.

Key point: Read the direction of each primer's free 3′ end.

Q10 / B primary versus secondary productivity

Answer: (B) 900 kJ m^{-2} day^{-1}.

Given: Gross productivity includes plant respiration.

Formula/Principle: NPP=GPP−autotrophic respiration.

Step 1: 1400−500=900.

Why: The herbivore assimilation is a later transfer of produced plant matter, not a term subtracted from the definition of plant NPP. Consumer assimilation may later affect ecosystem energy transfers, but it does not enter the gross-minus-plant-respiration definition used here.

Why not the others: (A) is herbivore assimilation; (C) adds it; (D) adds respiration.

Key point: Separate production from consumer uptake.

Q11 / C transcription regulation

Answer: (C) Operator escape permits transcription.

Why: The operator is a DNA binding site; eliminating repressor binding removes this negative gate while retaining functional structural genes and repressor protein. Other regulatory factors can still modulate output, so the inference is permission for constitutive expression under the model.

Why not the others: (A) predicts opposite effect; (B) ignores DNA-level transcription control; (D) confuses site mutation with protein deletion.

Key point: Distinguish a regulator from its binding site.

Q12 / D genetic-code reading

Answer: (D) Two.

Why: AUG codes methionine and initiates; GAA codes glutamate; UAA is a stop signal and does not add an amino acid. The reading frame is stipulated to avoid ambiguity from an upstream start.

Why not the others: (A) overlooks GAA; (B) counts stop as an amino acid; (C) counts nucleotides as codons.

Key point: A stop codon ends translation without coding a residue.

Q13 / D immune-related physiology

Answer: (D) High versus low TSH localises the fault.

Why: Low thyroid output relieves inhibitory feedback, so an intact pituitary should raise TSH in primary gland failure. A low TSH despite low thyroid hormone points upstream to deficient pituitary stimulation under the stated alternatives; a hypothalamic cause remains another upstream possibility.

Why not the others: (A) contradicts low hormone; (B) reverses the response; (C) is unsupported by the paired measurements.

Key point: Interpret a hormone together with its upstream controller.

Q14 / B chromosome versus DNA content

Answer: (B) Eight chromosomes and 4C DNA.

Given: Chromosome count tracks centromeres, and chromatids remain joined.

Formula/Principle: S phase doubles DNA mass but not chromosome number.

Step 1: G1 8 and 2C becomes metaphase 8 duplicated chromosomes and 4C.

Why: Each of the eight chromosomes now has two sister chromatids; the latter are not counted as separate chromosomes until centromeres divide. At anaphase sister centromeres separate, which is the later event that changes chromosome counting within the same cell.

Why not the others: (A) prematurely counts chromatids; (C) misses DNA duplication; (D) combines both errors.

Key point: Count centromeres separately from DNA amount.

Q15 / C gel fragment interpretation

Answer: (C) One cut yields the two lengths.

Why: Two Y fragments sum to the starting 900 bp and are compatible with one internal restriction site; uncut X yields the full-length band. Band positions provide fragment lengths, not quantitative mass or genotype without additional lane and intensity context.

Why not the others: (A) band positions alone do not measure total DNA mass; (B) uncut X does not prove a heterozygous genotype; (D) position alone does not establish fluorescence intensity.

Key point: Read fragment sizes, not hidden abundance.

Chemistry

Q16 / A VSEPR angle with lone pair

Answer: (A) NH_3 has the wider angle.

Why: F draws bonding electron density away from N, weakening N-centred bond-pair repulsions relative to the lone pair and compressing F–N–F. Both are trigonal pyramidal, but identical domain count does not imply identical angles.

Why not the others: (B) invokes atomic size alone; (C) overlooks electronegativity; (D) incorrectly removes the lone pair.

Key point: VSEPR angles depend on where bonding density lies.

Q17 / B ligand-field high-low spin

Answer: (B) Zero.

Why: Large octahedral splitting makes pairing in lower t_2g orbitals preferable to promotion into e_g. Six d electrons fill t_2g as three pairs, leaving e_g empty, so all spins are paired.

Why not the others: (A) is a weak-field d^6 case; (C) misallocates a pair; (D) ignores Pauli pairing and t_2g capacity.

Key point: Compare splitting to pairing energy before filling d levels.

Q18 / C Infer an unpaired-electron count from a magnetic moment

Answer: (C) Four unpaired electrons; count ratio four.

Given: The spin-only model is explicitly assumed, so orbital contributions are neglected.

Formula/Principle:

$$\mu/\mu_B = \sqrt{n(n+2)}.$$

Step 1:

$$n(n+2) = 24 \Rightarrow (n+6)(n-4) = 0.$$

The physical nonnegative integer solution is $n = 4$.

Step 2: For the reference ion, $m(m+2) = 3$, giving $m = 1$.

Step 3:

$$n/m = 4/1 = 4.$$

Why: Inferring the count requires solving the nonlinear spin-only relation rather than equating the magnetic moment to n. The negative algebraic root is inadmissible for a count of electrons.

Why not the others: (A) assumes moment proportional to electron count without using the formula; (B) fails $n(n+2) = 24$; (D) would predict $\sqrt{48}$, not $\sqrt{24}$, BM.

Key point: Invert the magnetic-moment relation and reject nonphysical roots.

Q19 / D orbital occupancy and energy

Answer: (D) Pauli exclusion.

Why: A complete set of four identical quantum numbers would specify an identical one-electron state for both electrons. Fermions cannot occupy that same state; opposite spins can share one spatial orbital. Energy degeneracy does not suspend this rule.

Why not the others: (A) concerns filling degenerate orbitals; (B) orders energies; (C) does not state the identical-state occupancy restriction.

Key point: Compare all four quantum numbers.

Q20 / A effective nuclear charge trend

Answer: (A) Both local exceptions are real.

Why: B's first 2p electron is easier to remove than a Be 2s electron despite B's extra proton. N has three singly occupied 2p orbitals; O introduces one paired 2p electron with extra repulsion, lowering its first ionisation energy relative to N. The general across-period increase needs both corrections.

Why not the others: (B) ignores subshell and pairing; (C) misassigns Be and N configurations; (D) reverses metallic character and half-filled stability.

Key point: Apply electron configuration before using a broad periodic trend.

Q21 / B Unknown initial gas composition from a cooled residual

Answer: (B) $50/3$ mL hydrogen.

Given: Oxygen is the residual gas, so hydrogen is completely consumed.

Formula/Principle: At equal temperature and pressure, $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ consumes one oxygen volume per two hydrogen volumes.

Step 1: Let initial hydrogen be x mL; initial oxygen is $30 - x$ mL.

Step 2:

$$(30 - x) - x/2 = 5.$$

Step 3:

$$3x/2 = 25 \Rightarrow x = 50/3 \text{ mL}.$$

Step 4: Initial oxygen is $40/3$ mL, of which $25/3$ mL reacts, leaving the observed 5 mL.

Why: Condensing water removes product vapour from the measured final gas volume, so the remaining volume is unused oxygen alone. Reconstructing both initial amounts from one total and one residual volume is a reverse material balance, not just a leftover calculation from known starting volumes.

Why not the others: (A) treats consumed gas as hydrogen alone; (C) ignores the measured residual; (D) assumes an equal initial composition without evidence.

Key point: Use residual identity and stoichiometry to reconstruct an unknown gas mixture.

Q22 / C resonance versus inductive acidity

Answer: (C) Chloroacetic acid.

Why: Electron withdrawal through sigma bonds stabilises the carboxylate conjugate base relative to unsubstituted acetate. Both conjugate bases have carboxylate resonance; the extra chlorine effect is principally inductive in this pair.

Why not the others: (A) methyl is donating; (B) ignores substituent effects; (D) claims an extra resonance topology absent here.

Key point: Compare conjugate-base stability under matched functional groups.

Q23 / D carbonyl addition selectivity

Answer: (D) Cyanide carbon.

Why: The nucleophilic carbon end forms the new C–C bond at electrophilic carbonyl carbon; the π electrons move onto oxygen, then protonation gives a hydroxyl group. The resulting cyanohydrin retains a nitrile C≡N bond.

Why not the others: (A) is delivered during protonation; (B) is the pre-existing carbonyl atom; (C) gives the wrong connectivity of the usual product.

Key point: Follow bond formation and protonation separately.

Q24 / A sodium versus neutral reagent response

Answer: (A) Alcohol.

Why: A protic O–H group can release H_2 with sodium; failure to evolve CO_2 with bicarbonate argues against the stronger carboxylic acid. Water solubility and absent carbonyl are consistent supporting clues, though not a unique structure.

Why not the others: (B) reacts with bicarbonate; (C) has carbonyl; (D) lacks suitable acidic O–H.

Key point: Use positive and negative controls together.

Q25 / B Directing-group strategy followed by diazonium replacement

Answer: (B) Nitrate before bromination, then convert the nitro position through amine and diazonium to chloride.

Step 1: Nitration gives nitrobenzene; NO_2 directs subsequent bromination predominantly meta.

Step 2: Selective reduction gives 3-bromoaniline while preserving the ring's C–Br bond.

Step 3: Cold diazotisation replaces NH_2 by the diazonium group at that same ring carbon; CuCl replaces that group by Cl.

Why: The meta relationship is established while the nitro substituent controls the electrophilic substitution. Later functional-group transformations preserve the two ring positions, so choosing the initial directing group is as important as the final replacement reagent.

Why not the others: (A) lets bromine direct nitration ortho/para; (C) creates strongly activated aniline before uncontrolled bromination; (D) does not selectively establish and preserve the requested meta bromine/chlorine pair.

Key point: Set positional selectivity before changing the directing group.

Q26 / C amino acid charge

Answer: (C) Zwitterion.

Why: Proton transfer gives an ammonium group and a carboxylate group carrying opposite charges. At the isoelectric point the average net charge is zero, though equilibrium includes minor other forms.

Why not the others: (A) ignores dominant internal ionisation; (B) is net positive; (D) misstates carboxylate charge.

Key point: Net neutral need not mean absence of charges.

Q27 / D Nonideal vapour pressure and mixing energetics

Answer: (D) Weaker unlike attractions and endothermic mixing.

Why: In the usual molecular interpretation, replacing strong like-like attractions by weaker unlike attractions makes escape to the vapour easier, explaining the higher pressure. Breaking the original attractions costs more energy than is recovered on making the weaker unlike contacts, so mixing is endothermic in this model. These are tendencies of the stated model, not a universal theorem for every composition of every nonideal mixture.

Why not the others: (A) describes the usual negative-deviation tendency; (B) combines stronger unlike contacts with the wrong energy consequence; (C) pairs the pressure-consistent weaker contacts with the opposite mixing-enthalpy sign.

Key point: Link vapour escape and mixing energetics to the same intermolecular comparison.

Q28 / A conductivity dilution

Answer: (A) Different normalisations give different trends.

Why: Dilution puts fewer charge carriers in each unit volume, lowering κ, while Λm=κ/c measures conductance per mole and can increase as ion mobility is less hindered. The trends need not be identical because their denominators differ.

Why not the others: (B) reverses dilution; (C) denies strong-electrolyte ionisation; (D) ignores units and concentration.

Key point: Distinguish per-volume from per-mole conductance.

Q29 / D integrated half-life relation

Answer: (D) 1/8.

Given: 60 min spans three half-lives.

Formula/Principle: Fraction remaining=(1/2)^(t/t_half).

Step 1: $(1/2)^3=1/8$.

Worked derivation:

$$N(60)/N_0 = (1/2)^{60/20} = (1/2)^3 = 1/8.$$

Why: Each equal half-life halves the amount remaining, not a fixed amount of the initial sample. Three successive halvings produce 1/2, 1/4, 1/8.

Why not the others: (A) counts two periods; (B) counts five; (C) counts four.

Key point: Exponential decay counts elapsed half-life intervals.

Q30 / C reaction quotient shift

Answer: (C) Left.

Why: Initially Kp=$2^2/1=4$. Immediately after adding NO_2, Qp=$4^2/1=16>$Kp, so net reaction consumes product until quotient falls back toward Kp. Temperature fixes Kp but does not force a disturbed mixture to remain at equilibrium.

Why not the others: (A) treats product addition as a right-driving disturbance; (B) confuses fixed Kp with the temporarily changed Qp; (D) interprets squaring the product pressure in the opposite direction.

Key point: Evaluate Q immediately after a disturbance.

Mathematics

Q31 / A hyperbola asymptote

Answer: (A) tan θ=4/3.

Given: a=3 and b=4.

Formula/Principle: Asymptotes of $x^2/a^2-y^2/b^2=1$ are $y=\pm(b/a)x$.

Step 1: Magnitude of slope is 4/3.

Worked derivation:

$$0=\frac{x^2}{9}-\frac{y^2}{16}\Rightarrow y=\pm\frac{4}{3}x, \qquad \tan\theta=\left|\frac{y}{x}\right|=\frac{4}{3}.$$

Why: Asymptote slopes arise by setting the leading quadratic homogeneous part to zero, not from focus distance. The asymptotes are straight lines through the hyperbola's centre, and their rise/run is set by b/a rather than a/b.

Why not the others: (B) inverts slope; (C) uses b/c; (D) uses c/b.

Key point: Read slope b/a for transverse x-axis.

Q32 / B Diameter midpoint and tangent construction

Answer: (B) $4x+3y=0$.

Given: $A=(4,0)$, $B=(0,3)$, so the centre is $C=(2,3/2)$.

Formula/Principle: A tangent is perpendicular to the radius at the point of contact.

Step 1:

$$CA^2=2^2+(3/2)^2=25/4=CO^2,$$

so the origin does lie on the circle.

Step 2: The radius CO has direction $(2,3/2)$. A tangent through O therefore has equation $2x+(3/2)y=0$.

Step 3: Multiply by two to obtain $4x+3y=0$.

Worked derivation:

$$(x-2)^2+(y-3/2)^2=25/4, \qquad \text{tangent at O: } 4x+3y=0.$$

Why: The given intercept line is the diameter line, not the tangent at O. First locating the midpoint of the diameter identifies the radius direction needed for tangency.

Why not the others: (A) reuses the diameter's normal; (C) reverses the required sign; (D) assumes equal intercept lengths.

Key point: Recover the centre before constructing a tangent.

Q33 / C parameterised differentiability

Answer: (C) −5/3.

Given: Continuity gives a=b+c; equal one-sided slopes give 2a=b; also 2b+c=5.

Formula/Principle: Solve these three linear constraints.

Step 1: c=a−2a=−a; 4a−a=5, so a=5/3, c=−5/3.

Worked derivation:

$$a = b + c,\ b = 2a,\ 2b + c = 5 \Rightarrow 3a = 5, \quad c = -\frac{5}{3}.$$

Why: Differentiability requires both matching value and matching derivative; matching only values leaves a free parameter. The third value f(2)=5 fixes the remaining scale after the first two conditions, leading to a negative intercept c.

Why not the others: (A) assumes a=1; (B) uses positive a; (D) treats f(2) as c.

Key point: Use continuity and equal one-sided derivatives.

Q34 / C Definite integration by parts

Answer: (C) 1.

Given: The integrand is a product of a linear factor and an exponential.

Formula/Principle: Integration by parts uses $u = x,\ dv = e^x dx$.

Step 1:

$$\int xe^x dx = xe^x - \int e^x dx = (x-1)e^x.$$

Step 2:

$$I = [(x-1)e^x]_0^1 = 0 - (-1) = 1.$$

Worked derivation:

$$I = [xe^x]_0^1 - [e^x]_0^1 = e - (e-1) = 1.$$

Why: Differentiating the algebraic factor reduces its degree and leaves an exponential with a simple antiderivative. The lower endpoint contributes minus one and cannot be discarded.

Why not the others: (A) integrates e^x alone; (B) retains only the boundary product; (D) drops the lower endpoint of the final primitive.

Key point: Evaluate both endpoints after integration by parts.

Q35 / D composition domain

Answer: (D) Exclude ±√3.

Worked derivation:

$$4 - x^2 \geq 0 \Rightarrow -2 \leq x \leq 2; \quad f(x) \neq 1 \Rightarrow \sqrt{4-x^2} \neq 1 \Rightarrow x \neq \pm\sqrt{3}.$$

Why: The outer function is undefined when its input equals one, so those two real preimages must be removed from the inner function's allowed interval. Endpoints give f=0 and are permitted.

Why not the others: (A) misses poles; (B) wrongly loses endpoints; (C) solves f(x)=1 incorrectly.

Key point: Impose both inner and outer domains.

Q36 / C determinant row transform

Answer: (C) 10.

Given: Adding a multiple of a row to another preserves determinant.

Formula/Principle: One row interchange reverses determinant sign.

Step 1: New determinant=−D=−10.

Worked derivation:

$$D' = \det(r_3, r_2 + 3r_1, r_1) = -D = -10 \Rightarrow D = 10.$$

Why: The addition does not contribute a factor three; only the swap changes the determinant. Reversing the sequence of operations gives the same sign conclusion only if the row addition is tracked on the correct current rows.

Why not the others: (A) multiplies by three and flips; (B) ignores swap; (D) multiplies by three.

Key point: Apply row-operation effects sequentially.

Q37 / B Bayes update from two conditionally independent tests

Answer: (B) 16/97.

Given: $P(X) = 0.1$, $P(+|X) = 0.8$, $P(+|\bar{X}) = 0.1$.

Formula/Principle: Use conditional independence within each type before applying Bayes' theorem.

Step 1:

$$P(+,-|X) = 0.8(0.2) = 0.16, \quad P(+,-|\bar{X}) = 0.1(0.9) = 0.09.$$

Step 2:

$$P(X,+,-) = 0.1(0.16) = 0.016, \quad P(\bar{X},+,-) = 0.9(0.09) = 0.081.$$

Step 3:

$$P(X|+,-) = \frac{0.016}{0.016 + 0.081} = \frac{16}{97}.$$

Worked derivation:

$$P(X|+,-) = \frac{0.1 \cdot 0.8 \cdot 0.2}{0.1 \cdot 0.8 \cdot 0.2 + 0.9 \cdot 0.1 \cdot 0.9} = \frac{16}{97}.$$

Why: Test repetitions are independent only after the hidden type is fixed; their unconditional outcomes cannot simply be multiplied. Both the positive and the negative result update the type probability, so ignoring either loses evidence.

Why not the others: (A) uses the first positive only; (C) ignores the prior and the negative result; (D) ignores both observations.

Key point: Compute the likelihood of the whole observed sequence within each hypothesis.

Q38 / D middle-term ratio

Answer: (D) 8/3.

Given: Coefficient x^3=C(6,3)2^3=160; x^4=C(6,4)2^2=60.

Formula/Principle: Binomial theorem with x powers tracked by chosen k.

Step 1: 160/60=8/3.

Worked derivation:

$$\frac{\binom{6}{3}2^3}{\binom{6}{4}2^2} = \frac{20 \cdot 8}{15 \cdot 4} = \frac{8}{3}.$$

Why: The exponent of 2 is the complement of the exponent of x, so adjacent coefficients contain both binomial and power factors. Taking only the ratio of binomial coefficients loses the extra factor of two from the different powers of the constant term.

Why not the others: (A) uses only powers; (B) uses binomial ratio alone; (C) rounds.

Key point: Track both factors in each term.

Q39 / C Compare AP and GP sums with shared endpoints

Answer: (C) 1542.

Given: The GP sum is $2^n - 1 = 1023$.

Formula/Principle: Determine the common term count and endpoints before using the AP endpoint-sum formula.

Step 1:

$$2^n = 1024 = 2^{10} \Rightarrow n = 10.$$

Step 2: The GP's last term is $2^{n-1} = 512$, so the AP also runs from 1 to 512 over ten terms.

Step 3:

$$S_{\mathsf{AP}} = \frac{10}{2}(1 + 512) = 2565.$$

Step 4:

$$S_{\mathsf{AP}} - S_{\mathsf{GP}} = 2565 - 1023 = 1542.$$

Worked derivation:

$$n = 10, \quad l = 512, \quad \Delta S = \tfrac{10}{2}(513) - 1023 = 1542.$$

Why: Equal endpoints do not imply equal sums when the intervening terms follow different progression rules. The GP sum first fixes the number of terms, which is required for the AP sum; all sums here are finite and explicitly within the listed progression syllabus.

Why not the others: (A) reports the original GP sum; (B) reports the AP sum without subtracting; (D) doubles the GP sum without using the AP endpoints.

Key point: Recover shared constraints before comparing two different progressions.

Q40 / D continuity parameter

Answer: (D) 2.

Given: k is a constant and sin(kx)/(kx) tends to 1 as x tends to zero.

Formula/Principle: Continuity requires $\lim_{x\to 0} f(x) = f(0)$.

Step 1: sin(kx)/x=k[sin(kx)/(kx)] tends to k (including k=0 by direct evaluation).

Worked derivation:

$$\lim_{x\to 0} \frac{\sin(kx)}{x} = k, \qquad k = f(0) = 2.$$

Why: The parameter controls the limiting slope of sine near zero; cancellation of an algebraic factor is not the method here. The same limit holds for negative k by the scaled sine limit, but only k=2 matches the explicitly assigned value at zero.

Why not the others: (A) gives limit −2; (B) gives zero; (C) gives one.

Key point: Use the standard trigonometric limit with its scale factor.

Q41 / B foot of perpendicular

Answer: (B) (6/5,8/5).

Given: Normal vector n=(3,4), n·P−10=15.

Formula/Principle: Foot H=P−[(n·P−10)/norm(n)²]n.

Step 1: Subtract (15/25)(3,4)=(9/5,12/5) from P, giving (6/5,8/5).

Worked derivation:

$$H = (3,4) - \frac{15}{25}(3,4) = \left(\frac{6}{5}, \frac{8}{5}\right).$$

Why: The correction vector is along the line's normal; its endpoint, not the correction itself, is the foot. Because the point P itself does not lie on the line, returning the displacement vector would confuse a direction-and-length with a coordinate.

Why not the others: (A) projects onto y-axis; (C) reports correction; (D) fails line equation.

Key point: Subtract the normal displacement from P.

Q42 / B path count avoiding a point

Answer: (B) 42.

Given: Paths via A: C(3,1)C(5,2)=30; via B: C(5,2)C(3,1)=30. Paths via both: C(3,1)C(2,0)C(3,1)=9.

Formula/Principle: Exactly one=through A+through B−2(through both).

Step 1: 30+30−18=42.

Worked derivation:

$$N_{\text{exactly one}} = \binom{3}{1}\binom{5}{2} + \binom{5}{2}\binom{3}{1} - 2\binom{3}{1}\binom{2}{0}\binom{3}{1} = 42.$$

Why: A path visiting both checkpoints appears in each individual through count and must be removed twice to satisfy exactly one. The order A then B is possible but B then A is impossible with only right/up steps, which makes the double-visit product unambiguous.

Why not the others: (A) keeps double visits twice; (C) removes them once; (D) counts only double visits.

Key point: Correct double-counting for an exclusive condition.

Q43 / A conjugate polynomial condition

Answer: (A) The conjugate is a root.

Worked derivation:

$$P(2+i) = 0, \quad P(2-i) = \overline{P(2+i)} = 0 \qquad (P \in \mathbb{R}[x]).$$

Why: Real coefficients commute with complex conjugation, forcing nonreal roots to occur in conjugate pairs. The remaining roots or leading coefficient are unrestricted.

Why not the others: (B) is not conjugation; (C) allows higher degree; (D) depends on other roots and leading coefficient.

Key point: Conjugate-root theorem requires real coefficients.

Q44 / A Initial-value solution and area after proving curve ordering

Answer: (A) $(4e - e^2 - 1)/4$.

Given: Use the homogeneous substitution $y = xv$, so $y' = v + xv'$.

Formula/Principle: Solve the initial-value equation, establish which curve is above, then integrate the nonnegative difference.

Step 1:

$$v + xv' = 1 + v \Rightarrow v' = 1/x, \quad v = \ln x + C.$$

Since y(1)=0, $C = 0$, hence $y = x \ln x$.

Step 2: Let $h(x) = x \ln x - x + 1$. Then $h(1) = 0$ and $h'(x) = \ln x \geq 0$ on the interval, so the solution lies above the line.

Step 3:

$$\mathcal{A} = \int_1^e (x \ln x - x + 1)\, dx.$$

Step 4: Integration by parts gives

$$\mathcal{A} = \left[\frac{x^2}{2}\ln x - \frac{3x^2}{4} + x\right]_1^e = e - \frac{e^2}{4} - \frac{1}{4}.$$

Worked derivation:

$$\mathcal{A} = (4e - e^2 - 1)/4 > 0.$$

Why: An area integral requires the correct vertical ordering, so solving the differential equation alone is insufficient. The derivative of the curve difference establishes that no interior sign split is needed before evaluating the definite integral.

Why not the others: (B) integrates only part of the solution expression; (C) gives the area under $x \ln x$ without subtracting the line; (D) ignores the varying curve separation.

Key point: Solve, compare and integrate in that order.

Q45 / A compound-angle identity

Answer: (A) −3/4.

Given: Square the given identity.

Formula/Principle: $(\sin\theta+\cos\theta)^2=1+\sin 2\theta$.

Step 1: $1/4=1+\sin 2\theta$.

Worked derivation:

$$\sin 2\theta = (\sin\theta + \cos\theta)^2 - 1 = \frac{1}{4} - 1 = -\frac{3}{4}.$$

Why: The stipulated sum fixes sin2θ even though many distinct θ satisfy it; squaring is sufficient for this requested invariant. The real angle need not be found explicitly; the given sum already determines the double-angle invariant via the Pythagorean identity.

Why not the others: (B) treats product incorrectly; (C) omits −1; (D) loses sign.

Key point: Squared sum directly contains double-angle sine.

Physics

Q46 / B dimension of coefficient

Answer: (B) L T^{-2} and L.

Worked derivation:

$$[At^2] = L \Rightarrow [A] = LT^{-2}; \quad [\sin(\omega t)] = 1 \Rightarrow [B] = L.$$

Why: Arguments of trig functions must be dimensionless, hence ωt has no dimension; each additive term of x separately has dimension length. B must carry length because it multiplies a dimensionless sine, even though the oscillation frequency has units of inverse time.

Why not the others: (A) assigns velocity to A; (C) ignores time factor; (D) treats sine as a time quantity.

Key point: Check each sum term and trig argument.

Q47 / C constraint acceleration

Answer: (C) g/3.

Given: Net driving force is (2m−m)g=mg; total accelerated mass 3m.

Formula/Principle: Combine two Newton equations, eliminating tension.

Step 1: a=mg/(3m)=g/3.

Worked derivation:

$$2mg - T = 2ma, \quad T - mg = ma \Rightarrow mg = 3ma \Rightarrow a = \frac{g}{3}.$$

Why: Tension is internal to the two-block system; subtract weights but add masses to compute acceleration. If only the heavier block's mass were used in the denominator, the shared string would be accelerating an omitted second block.

Why not the others: (A) uses an incorrect total mass; (B) overcounts inertia; (D) treats the net driving weight as 2mg.

Key point: Keep drive and inertia distinct.

Q48 / D series wire extension

Answer: (D) 2.

Given: Series tension F is the same in both wires.

Formula/Principle: Extension ΔL=FL/(YA).

Step 1: Halving area doubles extension at fixed F,L,Y.

Worked derivation:

$$\frac{\Delta L_A}{\Delta L_{2A}} = \frac{FL/(YA)}{FL/(2YA)} = 2.$$

Why: A series connection shares force, not extension; the thinner wire has twice the strain under equal material and length. This ratio remains unchanged when both wires are stretched more strongly, as long as they stay in the linear elastic regime.

Why not the others: (A) reverses ratio; (B) assumes equal extension; (C) squares area ratio.

Key point: Force is common through series wires.

Q49 / A thermal-expansion constraint

Answer: (A) YαΔT.

Given: Free thermal strain would be +αΔT; actual total strain is zero.

Formula/Principle: Mechanical compressive strain must be −αΔT, with magnitude stress/Y.

Step 1: |σ|/Y=αΔT.

Worked derivation:

$$0 = \epsilon_{\text{mech}} + \alpha\Delta T \Rightarrow \epsilon_{\text{mech}} = -\alpha\Delta T, \quad |\sigma| = Y\alpha\Delta T.$$

Why: Constraints prevent geometric expansion but produce internal compression rather than eliminating thermal tendency. The Young modulus relates the compensating mechanical strain to stress, so a higher modulus generates a larger force under the same restraint.

Why not the others: (B) divides by α incorrectly; (C) inverts Y rather than multiplying it; (D) mistakes fixed length for absence of thermal stress.

Key point: Add thermal and mechanical strains.

Q50 / B equipartition heat capacity

Answer: (B) 7R/2.

Given: Ideal-gas Mayer relation CP−CV=R.

Formula/Principle: CV=5R/2 under the stated active degrees of freedom.

Step 1: CP=5R/2+R=7R/2.

Worked derivation:

$$C_P = C_V + R = \frac{5R}{2} + R = \frac{7R}{2}.$$

Why: The extra R accounts for pressure-volume work during constant-pressure heating, not an additional internal mode. The stipulated frozen vibrational modes prevent adding another thermal degree-of-freedom term to the internal energy.

Why not the others: (A) removes R; (C) equates heat capacities; (D) adds two R.

Key point: Keep active modes and process work separate.

Q51 / C velocity-acceleration relation

Answer: (C) North, toward centre.

Worked derivation:

$$\vec{a} = \frac{v^2}{r}\,\hat{\mathbf{r}}_{\text{inward}}, \quad |\vec{v}| = \text{constant but } d\vec{v}/dt \neq 0.$$

Why: The velocity vector changes direction despite unchanged magnitude, producing inward centripetal acceleration. Its direction is fixed by the centre's position, not by motion direction.

Why not the others: (A) east is the instantaneous velocity direction, not acceleration; (B) west is the opposite tangent; (D) confuses unchanging speed with unchanging velocity vector.

Key point: Acceleration describes change of vector velocity.

Q52 / D reflection phase at boundary

Answer: (D) Inverted pulse.

Worked derivation:

$$r_{\text{amplitude}} \to -1 \quad \text{as the load approaches a fixed boundary;} \quad \Delta\phi = \pi.$$

Why: To keep the junction displacement small in a fixed-end limit, incident and reflected displacements nearly cancel at the boundary; that demands opposite signs. The limiting reflected pulse travels back in the lighter string with reversed displacement even though its direction of propagation also reverses.

Why not the others: (A) describes free-end reflection; (B) ignores impedance mismatch; (C) wrongly extends boundary condition to entire string.

Key point: Use displacement boundary condition to infer phase.

Q53 / A flux of multi-charge surface

Answer: (A) $-2q/\varepsilon_0$.

Given: Enclosed net charge is $+q-3q=-2q$.

Formula/Principle: Gauss law $\Phi E = Q_{enc}/\varepsilon_0$.

Step 1: Exclude external $+5q$ from enclosed total.

Worked derivation:

$$\Phi_E = \frac{q - 3q}{\epsilon_0} = -\frac{2q}{\epsilon_0}.$$

Why: External charge alters local field crossing parts of the sphere, but its inward and outward contributions cancel in the net closed-surface flux. The sign is negative because the enclosed negative charge has greater magnitude than the enclosed positive charge.

Why not the others: (B) takes magnitude of one charge; (C) uses outside charge; (D) omits $+q$.

Key point: Closed flux depends on enclosed net charge.

Q54 / A resistor network node reduction

Answer: (A) R.

Given: Series A–B–C branch totals 2R; direct A–C branch is 2R.

Formula/Principle: Two equal 2R branches in parallel give R.

Step 1: $R_{eq} = (2R)(2R)/(2R+2R) = R$.

Worked derivation:

$$R_{\text{eq}} = (R + R) \parallel 2R = \left(\frac{1}{2R} + \frac{1}{2R}\right)^{-1} = R.$$

Why: The intermediate B node has no other branch, allowing its two resistors in series; both resulting paths share endpoints A and C. This reduction remains valid because B is not shorted to A or C by another ideal wire, as the diagram explicitly shows.

Why not the others: (B) treats all three parallel; (C) ignores parallel path; (D) adds both branches.

Key point: Identify nodes before combining resistors.

Q55 / B Lenz direction under flux change

Answer: (B) Counterclockwise.

Worked derivation:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}; \quad \text{increasing inward flux} \Rightarrow \text{induced outward field.}$$

Why: Lenz's law opposes the change in flux, not necessarily the original field. By right-hand grip rule, a counterclockwise loop current makes an outward magnetic field as viewed face-on.

Why not the others: (A) aids change; (C) mismatches current and field; (D) ignores changing magnitude.

Key point: Determine opposing field before current direction.

Q56 / C inductor-capacitor voltage cancellation

Answer: (C) 10 Ω.

Given: Net reactance XL−XC=8 Ω.

Formula/Principle: Series impedance magnitude Z=√[R²+(XL−XC)²].

Step 1: √(36+64)=10 Ω.

Worked derivation:

$$Z = \sqrt{6^2 + (10-2)^2} = \sqrt{100} = 10\ \Omega.$$

Why: Inductive and capacitive voltages oppose in phase; their reactances subtract before combining quadratically with resistance. At the selected frequency the inductor dominates the capacitor by 8 Ω, but that 8 Ω is reactive and cannot be added arithmetically to 6 Ω.

Why not the others: (A) ignores resistance; (B) adds R and reactance; (D) adds all impedances.

Key point: Reactances subtract; resistance is perpendicular in phasor space.

Q57 / B lens-mirror combination

Answer: (B) 30 cm on the original object side.

Given: The thin converging lens has $f = 20$ cm and initial real-object distance 30 cm; the plane mirror is at the first image plane.

Formula/Principle: Apply the thin-lens conjugate relation to each traversal, measuring real-object and real-image distance magnitudes along that traversal's light direction.

Step 1:

$$\frac{1}{v_1} = \frac{1}{20} - \frac{1}{30} = \frac{1}{60} \Rightarrow v_1 = 60 \text{ cm}.$$

Step 2: Reflection at the mirror plane produces a returning bundle diverging from that same axial point, so the return object is 60 cm from the lens on the mirror side.

Step 3:

$$\frac{1}{v_2} = \frac{1}{20} - \frac{1}{60} = \frac{1}{30} \Rightarrow v_2 = 30 \text{ cm}.$$

Step 4: Return light travels toward the original object side; its real image is therefore there, not beyond the mirror.

Worked derivation:

$$30 \xrightarrow{\text{lens}} 60 \xrightarrow{\text{mirror, return lens}} 30 \text{ cm on original side.}$$

Why: The image-plane point is the object for the return pass, so the second calculation uses a different object distance from the first. Individual oblique rays need not retrace their incident lines for the bundle to form the same conjugate axial point.

Why not the others: (A) incorrectly collapses a finite-distance conjugate to the lens plane; (C) reports focal length rather than the return image distance; (D) leaves the image on the wrong side for the returning real image.

Key point: Define the object and light direction again on every optical traversal.

Q58 / D Infer amplitudes from contrast then apply intensity attenuation

Answer: (D) 8/17.

Given: Let the original positive amplitudes be a>b; intensities are proportional to amplitude squared.

Formula/Principle:

$$I_{\max}/I_{\min} = (a+b)^2/(a-b)^2, \qquad V = \frac{2ab}{a^2+b^2}.$$

Step 1:

$$(a+b)/(a-b) = 3 \Rightarrow a = 2b.$$

Step 2: Reducing the weaker intensity fourfold changes its amplitude from b to b/2, so the new amplitude ratio is $a : (b/2) = 4 : 1$.

Step 3:

$$V_f = \frac{2(4)(1)}{4^2+1^2} = \frac{8}{17}.$$

Worked derivation:

$$I_{\max,f} : I_{\min,f} = 5^2 : 3^2 = 25 : 9, \quad V_f = (25-9)/(25+9) = 8/17.$$

Why: The initial contrast determines an amplitude ratio, while the later attenuation is specified as an intensity change. Taking its square root before updating the amplitudes is essential; coherence alone does not guarantee unit visibility.

Why not the others: (A) leaves the initial visibility unchanged; (B) treats attenuation as an amplitude ratio directly; (C) reports the attenuator's intensity factor rather than fringe contrast.

Key point: Move carefully between intensity ratios, amplitude ratios and visibility.

Q59 / A de Broglie acceleration comparison

Answer: (A) √(mp/me).

Given: Each gains kinetic energy eV of equal magnitude.

Formula/Principle: p=√(2m eV) and λ=h/p.

Step 1: λ∝1/√m at fixed V and charge magnitude.

Worked derivation:

$$\frac{\lambda_e}{\lambda_p} = \frac{h/\sqrt{2m_e eV}}{h/\sqrt{2m_p eV}} = \sqrt{\frac{m_p}{m_e}}.$$

Why: Equal energy does not mean equal momentum; the lighter electron acquires smaller momentum and longer wavelength. The nonrelativistic condition matters because at sufficiently high accelerating voltage the momentum-energy relation would no longer be √(2meV).

Why not the others: (B) inverts; (C) omits square root; (D) assumes equal mass.

Key point: Compare momentum at equal kinetic energy.

Q60 / D diode I-V inference

Answer: (D) +0.8 V, high forward current.

Worked derivation:

$$V = +0.8 \text{ V} > 0.6 \text{ V} \Rightarrow \text{point lies on steep positive-current branch.}$$

Why: The graph itself locates the approximate turn-on region; at +0.8 V the forward-bias branch has risen markedly. A small reverse bias does not produce large positive current.

Why not the others: (A) small reverse bias does not give large forward current; (B) +0.2 V is below the depicted rise; (C) −0.8 V remains on the small reverse-current branch shown.

Key point: Read operating regime from the labelled plot.